

DBMS – PAYMENT MANAGEMENT

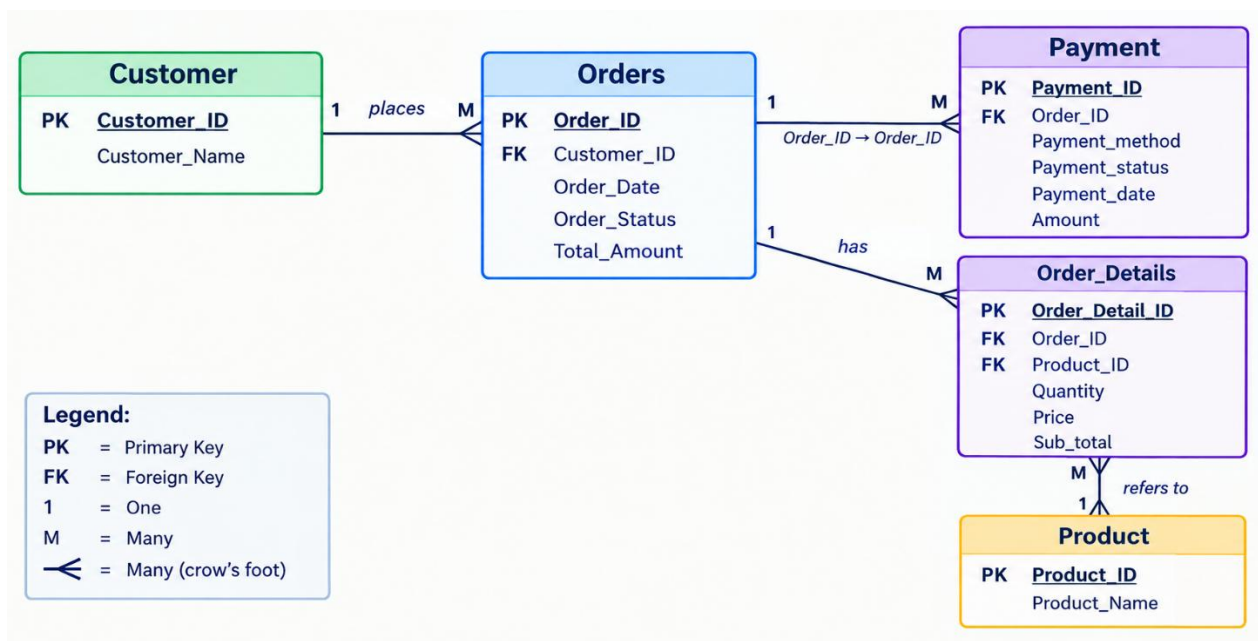
Project Title: Big Basket E-commerce Database Management System

1. Database Design

1.1 Payment Table

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_method	-
Payment_status	Check
Payment_date	-
Amount	-

2. Primary and Foreign Key Relationship



3. Table Creation

3.1 Create Payment Table

```
CREATE TABLE Payment (  
    Payment_ID INT PRIMARY KEY,  
    Order_ID INT,  
    Payment_method VARCHAR(50),  
    Payment_status VARCHAR(20) CHECK (Payment_status IN ('Pending', 'Paid', 'Failed', 'Refunded')),  
    Payment_date DATE,  
    Amount DECIMAL(10,2),  
    FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
);  
Table created .
```

4. Data Insertion

4.1 Payment Records

```
INSERT INTO Payment  
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)  
VALUES  
(1, 1001, 'UPI', 'Paid', TO_DATE('21-09-2026','DD-MM-YYYY'), 900);  
1 row created.  
  
INSERT INTO Payment  
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)  
VALUES  
(2, 1002, 'Credit Card', 'Paid', TO_DATE('20-09-2026','DD-MM-YYYY'), 1250);  
1 row created.  
  
INSERT INTO Payment  
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)  
VALUES  
(3, 1003, 'Debit Card', 'Paid', TO_DATE('19-09-2026','DD-MM-YYYY'), 1800);  
1 row created.  
  
INSERT INTO Payment  
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)  
VALUES  
(4, 1004, 'UPI', 'Paid', TO_DATE('18-09-2026','DD-MM-YYYY'), 750);  
1 row created.
```

```

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(5, 1005, 'Cash', 'Refunded', TO_DATE('17-09-2026','DD-MM-YYYY'), 500);
1 row created.

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(6, 1006, 'UPI', 'Pending', TO_DATE('16-09-2026','DD-MM-YYYY'), 850);
1 row created.

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(7, 1007, 'Credit Card', 'Refunded', TO_DATE('15-09-2026','DD-MM-YYYY'), 1200);
1 row created.

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(8, 1008, 'Debit Card', 'Paid', TO_DATE('04-09-2026','DD-MM-YYYY'), 1500);
1 row created.

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(9, 1009, 'Net Banking', 'Paid', TO_DATE('09-09-2026','DD-MM-YYYY'), 2500);
1 row created.

```

```

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(10, 1010, 'Credit Card', 'Paid', TO_DATE('20-09-2026','DD-MM-YYYY'), 28000);
1 row created.

INSERT INTO Payment
(Payment_ID, Order_ID, Payment_method, Payment_status, Payment_date, Amount)
VALUES
(11, 1006, 'Credit Card', 'Failed',
TO_DATE('16-09-2026','DD-MM-YYYY'), 850);

```

5. Data Retrieval

5.1 Display Payment

```

Select * from Payment;

```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
1	1001	UPI	Paid	21-SEP-26	900
2	1002	Credit Card	Paid	20-SEP-26	1250
3	1003	Debit Card	Paid	19-SEP-26	1800
4	1004	UPI	Paid	18-SEP-26	750
5	1005	Cash	Refunded	17-SEP-26	500
6	1006	UPI	Pending	16-SEP-26	850
7	1007	Credit Card	Refunded	15-SEP-26	1200
8	1008	Debit Card	Paid	04-SEP-26	1500
9	1009	Net Banking	Paid	09-SEP-26	2500
10	1010	Credit Card	Paid	20-SEP-26	28000
11	1006	Credit Card	Failed	16-SEP-26	850

```

11 rows selected.

```

6. Payment Modification

6.1 Update Payment Status

```
Update Payment
SET Payment_Status='Paid'
Where Order_ID=1005;
1 row updated.

Select *
From Payment
Where Order_ID=1005;
PAYMENT_ID  ORDER_ID  PAYMENT_METHOD
-----
PAYMENT_STATUS  PAYMENT_D  AMOUNT
-----
Paid           5          1005 Cash
              17-SEP-26          500
```

7. Manage Payment Transactions

7.1 Display Paid Payments

```
Select *
from Payment
Where Payment_Status='Paid';
PAYMENT_ID  ORDER_ID  PAYMENT_METHOD
-----
PAYMENT_STATUS  PAYMENT_D  AMOUNT
-----
Paid           1          1001 UPI
              21-SEP-26          900
Paid           2          1002 Credit Card
              20-SEP-26         1250
Paid           3          1003 Debit Card
              19-SEP-26         1800
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD

PAYMENT_STATUS	PAYMENT_D	AMOUNT

Paid	4 1004 UPI	750
	18-SEP-26	
Paid	8 1008 Debit Card	1500
	04-SEP-26	
Paid	9 1009 Net Banking	2500
	09-SEP-26	
PAYMENT_ID	ORDER_ID	PAYMENT_METHOD

PAYMENT_STATUS	PAYMENT_D	AMOUNT

Paid	10 1010 Credit Card	28000
	20-SEP-26	
7 rows selected.		

7.2 Display Failed Payments

```
Select *
from Payment
Where Payment_Status='Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD

PAYMENT_STATUS	PAYMENT_D	AMOUNT

Failed	11 1006 Credit Card	850
	16-SEP-26	

7.3 Display Pending Payments

```
Select *
from Payment
Where Payment_Status='Pending';
```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD

PAYMENT_STATUS	PAYMENT_D	AMOUNT

Pending	6 1006 UPI	850
	16-SEP-26	

7.4 Display Refunded Payments

```

Select *
from Payment
Where Payment_Status='Refunded';

```

PAYMENT_ID	ORDER_ID	PAYMENT_METHOD	PAYMENT_STATUS	PAYMENT_D	AMOUNT
5	1005	Cash	Refunded	17-SEP-26	500
7	1007	Credit Card	Refunded	15-SEP-26	1200

8. Analyse Payment Method

```

SELECT
    Payment_method,
    COUNT(*) AS Total_Payments,
    SUM(Amount) AS Total_Amount
FROM Payment
GROUP BY Payment_method
ORDER BY Total_Payments DESC;

```

PAYMENT_METHOD	TOTAL_PAYMENTS	TOTAL_AMOUNT
Credit Card	4	31300
UPI	3	2500
Debit Card	2	3300
Cash	1	500
Net Banking	1	2500

9. Payment Transaction Report

```

SELECT
    p.Payment_ID,
    o.Order_ID,
    c.Customer_ID,
    c.Customer_Name,
    p.Payment_method,
    p.Payment_status,
    p.Payment_date,
    p.Amount
FROM Payment p
JOIN Orders o
    ON p.Order_ID = o.Order_ID
JOIN Customer c
    ON o.Customer_ID = c.Customer_ID
ORDER BY p.Payment_date;

```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD			PAYMENT_STATUS	

PAYMENT_D	AMOUNT			

	8	1008	101	
Arun Kumar				
Debit Card			Paid	
04-SEP-26	1500			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD			PAYMENT_STATUS	

PAYMENT_D	AMOUNT			

	9	1009	107	
Karan Mehta				
Net Banking			Paid	
09-SEP-26	2500			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD			PAYMENT_STATUS	

PAYMENT_D	AMOUNT			

	7	1007	106	
Anjali Das				
Credit Card			Refunded	
15-SEP-26	1200			
PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD			PAYMENT_STATUS	

PAYMENT_D	AMOUNT			

	6	1006	105	
Vikram Patel				
UPI			Pending	
16-SEP-26	850			

payment.sql > ...

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD				PAYMENT_STATUS

PAYMENT_D	AMOUNT			

11	1006	105		
Vikram Patel				
Credit Card				Failed
16-SEP-26	850			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD				PAYMENT_STATUS

PAYMENT_D	AMOUNT			

5	1005	104		
Sneha Reddy				
Cash				Paid
17-SEP-26	500			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD				PAYMENT_STATUS

PAYMENT_D	AMOUNT			

4	1004	101		
Arun Kumar				
UPI				Paid
18-SEP-26	750			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		

CUSTOMER_NAME				

PAYMENT_METHOD				PAYMENT_STATUS

PAYMENT_D	AMOUNT			

3	1003	103		
Rahul Singh				
Debit Card				Paid
19-SEP-26	1800			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		
CUSTOMER_NAME				
PAYMENT_METHOD			PAYMENT_STATUS	
PAYMENT_D	AMOUNT			
10	1010	109		
Rohan Gupta				
Credit Card			Paid	
20-SEP-26	28000			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		
CUSTOMER_NAME				
PAYMENT_METHOD			PAYMENT_STATUS	
PAYMENT_D	AMOUNT			
2	1002	102		
Priya Sharma				
Credit Card			Paid	
20-SEP-26	1250			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID		
CUSTOMER_NAME				
PAYMENT_METHOD			PAYMENT_STATUS	
PAYMENT_D	AMOUNT			
1	1001	101		
Arun Kumar				
UPI			Paid	
21-SEP-26	900			

11 rows selected.